

Dr. Shibiao WAN

Assistant Professor

Assistant Director for Bioinformatics and Systems Biology Core

University of Nebraska Medical Center

Address: DRC I 6014, 985805 Nebraska Medical Center, Omaha, NE 68198

Office: +1-402-559-6560

Official Website: <https://www.unmc.edu/genetics/faculty/bios/wan.html>

Lab Website: <https://wan-mlab.github.io/>

Email: swan@unmc.edu



Highlights

- **>14 years of experience** in bioinformatics, machine learning and computational biology
- **>60 publications** (including one book and >50 top-tier journal articles)
- **Google Scholar Citations: >1600**
- **10 active grants** supported by NIH/Office of the Director, NIH/NCI, NIH/NIGMS, ACS, NRI and other fundings
- **Editor-in-Chief** for *Current Proteomics*
- **Associate Editor** for *Heliyon (Cell Press)*
- **Editorial Board Member** for *Briefings in Functional Genomics (Oxford University Press)*
- **Editorial Board Member** for *BMC Bioinformatics (Springer Nature)*
- **Editorial Board Member** for *Discover Oncology (Springer Nature)*
- **Editorial Board Member** for *PeerJ Computer Science (PeerJ)*
- **Editorial Board Member** for *International Journal of Microbiology (Wiley)*
- **Editorial Board Member** for *BioMed Research International (Wiley)*
- **Editorial Board Member** for *Computational and Mathematical Methods (Wiley)*
- **Guest Associate Editor** for
 - (1) *Frontiers in Immunology*; (2) *Frontiers in Cell and Developmental Biology*; (3) *Frontiers in Pharmacology*;
 - (4) *Frontiers in Oncology*; (5) *Frontiers in Genetics*; (6) *Frontiers in Psychology*.
 - (7) *International Journal of Molecular Sciences*; (8) *Biology*; (9) *Genes*.
 - (10) *BMC Methods*; (11) *Discover Life*.
- **TPC Member** for >20 machine learning international conferences including *IEEE ICTAI*
- **Reviewer** for >50 journals including ***Nature Methods***; *Nature Communications*; *Nature Computational Science*; *Nucleic Acids Research*; *Genome Medicine*; *Cancer Research*; *Briefings in Bioinformatics*.
- **Top Reviewers Award** (top 1%) in *Cross-Field*, and *Biology and Biochemistry* awarded by Clarivate
- Developed **20 highly accessed bioinformatics tools/webservers**
- **New Investigator Award** by University of Nebraska Medical Center (2024)
- **FIRST Award** by Nebraska EPSCoR Program (2023)
- **Outstanding Young Alumni Award** by The Hong Kong Polytechnic University (2022)
- **IEEE Senior Member**

Curriculum Vitae

ACADEMIC EXPERIENCE

- 2024 - present *Co-Director of Bioinformatics Master's Program*, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2024 - present *Assistant Professor (courtesy)*, Department of Biostatistics, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2022 - present *Assistant Professor*, Department of Genetics, Cell Biology and Anatomy, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2022 - present *Assistant Director for Bioinformatics and Systems Biology Core*, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2019 - 2022 *Bioinformatics Research Scientist*, [St. Jude Children's Research Hospital](#), Memphis, TN, USA
- 2017 – 2019 *Postdoctoral Researcher*, [University of Pennsylvania](#), Philadelphia, PA, USA
- 2016 – 2017 *Postdoctoral Research Associate*, [Princeton University](#), Princeton, NJ, USA
- 2014 – 2016 *Postdoctoral Fellow*, [The Hong Kong Polytechnic University](#), Hong Kong SAR
- 2014 *Research Associate*, [The Hong Kong Polytechnic University](#), Hong Kong SAR
- 2013 *Visiting Scholar*, [Johns Hopkins University School of Medicine](#), MD, USA and [CBIL lab](#) of [Virginia Tech](#), VA, USA.

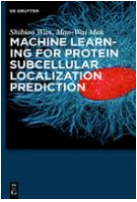
EDUCATION

- 2010 – 2014 *Ph.D.*, [The Hong Kong Polytechnic University](#), Hong Kong SAR
Dissertation Advisor: Dr. Man-Wai Mak
- 2006 – 2010 *B.Eng.*, [Wuhan University](#), Wuhan, China.
Thesis Advisor: Prof. Deshi Li

RESEARCH INTERESTS

- **Bioinformatics and Computational Biology:**
 - Single-Cell Analysis, Spatial Omics Analysis, Multi-Omics Analysis, Alternative Splicing.
 - NGS Data Analysis, Genomics, Epigenomics, Proteomics.
 - Cancer Research, Leukemia Subtyping, Cancer Disparities.
 - Intelligent Healthcare, Precision Medicine, Alzheimer's Disease, Parkinson's Disease.
 - Sequence Analysis, Antimicrobial Peptide Identification, Protein Subcellular Localization.
- **Machine Learning/Artificial Intelligence:**
 - Deep Learning, Transfer Learning, Interpretable Machine Learning.
 - Big Data Analytics, High-Dimensional Data Analysis, Dimension Reduction.
 - Unsupervised Learning, Supervised Learning, Data and Text Mining, Computational Modeling.
 - Semi-Supervised Learning, Transductive Learning, Kernel Learning, Ensemble Learning.
 - Multi-Label Classification, Discriminative Models, Regression Analysis, Optimization.

PUBLICATIONS



BOOK

1. **S. Wan** and M. W. Mak, “[Machine Learning for Protein Subcellular Localization Prediction](#)”, ISBN 978-1-5015-0150-0, published by **De Gruyter**, Germany, 2015.
2. **S. Wan**, Y. Fan, C. Jiang and S. Li, “[Bioinformatics and Machine Learning for Cancer Biology](#)”, published by **MDPI**, ISBN 978-3-0365-4814-2, Switzerland, 2022. (Edited Book)

JOURNALS (Impact Factor (IF) from Journal Citation Reports (JCR); *: corresponding author; #: co-first author)

3. B. Kennedy, M. Raza, S. Mirza, A. Rajan, F. Oruji, M. Storck, S. Lele, T. Reznicek, L. Li, J. Rowley, **S. Wan**, B. Mohapatra, H. Band, and V. Band, “ECD co-operates with ERBB2 to promote tumorigenesis through upregulation of unfolded protein response and glycolysis”, **bioRxiv**, 2025, 2025.01.28.635284.
4. J. Fang, S. Singh, B. Wells, Q. Wu, H. Jin, L. Janke, **S. Wan**, J. Steele, J. P. Connelly, A. Murphy, R. Wang, A. M. Davidoff, M. Ashcroft, S. M. Pruett-Miller, and J. Yang, “The Context-Dependent Epigenetic and Organogenesis Programs Determine 3D vs. 2D Cellular Fitness”, **eLife**, 2025, vol. 14, pp. RP101299.
5. G. Feng, H. Xu, **S. Wan**, H. Wang, X. Chen, R. Magan, Y. Han, Y. Wei, and H. Gu, “Twelve Practical Recommendations for Developing and Applying Clinical Predictive Models”, **The Innovation Medicine**, 2024, vol. 2, no. 4, pp. 100105.
6. J. Feng[#], M. Sun[#], C. Liu, W. Zhang, C. Xu, J. Wang, G. Wang, and **S. Wan***, “SAMP: Identifying Antimicrobial Peptides by an Ensemble Learning Model Based on Proportionalized Split Amino Acid Composition”, **Briefings in Functional Genomics**, 2024, vol. 23, no. 6, pp. 879-890.
7. Z. Xu, L. Li, R. Liu, M. Azzam, **S. Wan**, and J. Wang, “Functional Connectivity Alterations in Cocaine Use Disorder: Insights from the Triple Network Model and the Addictions Neuroclinical Assessment Framework”, **bioRxiv**, 2024, 2024.11.12.623073.
8. M. Azzam, Z. Xu, R. Liu, L. Li, K. M. Soh, K. B. Challagundla, **S. Wan**, and J. Wang*, “A review of artificial intelligence-based brain age estimation and its applications for related diseases”, **Briefings in Functional Genomics**, 2024, elae042.
9. H. Xiao, J. Wang*, and **S. Wan***, “WIMOAD: Weighted Integration of Multi-Omics Data for Alzheimer's Disease (AD) Diagnosis”, **bioRxiv**, 2024, 2024.09.25.614862.
10. L. Li, H. Xiao, X. Wu, Z. Tang, J. Khoury, J. Wang, and **S. Wan***, “RanBALL: An Ensemble Random Projection Model for Identifying Subtypes of B-Cell Acute Lymphoblastic Leukemia”, **bioRxiv**, 2024, 2024.09.24.614777.
11. H. Chen, Y. Xu, H. Lin, **S. Wan***, and L. Luo*, “A Prognostic Framework for Predicting Lung Signet Ring Cell Carcinoma via a Machine Learning Based Cox Proportional Hazard Model”, **Journal of Cancer Research and Clinical Oncology**, 2024, vol. 150, no. 364, pp. 1-15. (JCR Q2; Rank: 79/214; IF: 2.7)

Curriculum Vitae

12. L. Li[#], M. Sun[#], J. Wang, and **S. Wan***, "Multi-Omics Based Artificial Intelligence for Cancer Research", *Advances in Cancer Research*, 2024, vol. 163, pp. 303-356.
13. Z. Ahmed[#], **S. Wan**[#], F. Zhang[#], and W. Zhong[#], "Artificial Intelligence for Omics Data Analysis", *BMC Methods*, 2024, vol. 1, no. 1, 4.
14. H. Xiao, Y. Zou, J. Wang, and **S. Wan***, "A Review for Artificial Intelligence Based Protein Subcellular Localization", *Biomolecules*, 2024, vol. 14, no. 4, 409. (*JCR Q1; Rank: 70/285; IF: 5.5*)
15. J. Zeng, Y. Weng, T. Lai, L. Chen, Y. Li, Q. Huang, S. Zhong, **S. Wan***, and L. Luo*, "Procyanidin alleviates ferroptosis and inflammation of LPS-induced RAW264.7 cell via the Nrf2/HO-1 pathway", *Naunyn-Schmiedeberg's Archives of Pharmacology*, 2024, vol. 397, no. 6, pp. 4055-4067. (*JCR Q2; Rank: 114/278; IF: 3.6*)
16. M. Sun[#], L. Li[#], H. Xiao[#], J. Feng[#], J. Wang, and **S. Wan***, "Editorial: Bioinformatics Analysis of Omics Data for Biomarker Identification in Clinical Research, Volume II", *Frontiers in Genetics*, 2023, vol. 14, 1256468. (*JCR Q2; Rank: 48/176; IF: 4.599*)
17. H. J. Mallard, **S. Wan**, P. Nidhi, Y. D. Hanscom-Trofy, B. Mohapatra, N. T Woods, J. A. Lopez-Guerrero, A. Llombart-Bosch, I. Machado, K. Scotlandi, N. F. Kreiling, M. C. Perry, S. Mirza, D. W. Coulter, V. Band, H. Band, and G. Ghosal, "USP1 Expression Driven by EWS::FLI1 Transcription Factor Stabilizes Survivin and Mitigates Replication Stress in Ewing Sarcoma", *Molecular Cancer Research*, 2023, vol. 21, no. 11, pp. 1186-1204. (*JCR Q2; Rank: 72/241; IF: 5.2*)
18. Y. Liang, Z. Su, X. Mao, **S. Wan*** and L. Luo*, "Editorial: Ferroptosis as a Novel Therapeutic Target for Inflammation-Related Diseases", *Frontiers in Pharmacology*, 2023, vol. 14, article 1152326. (*JCR Q1; Rank: 50/279; IF: 5.988*)
19. W. Shi, S. Mirza, M. Kuss, B. Liu, A. Hartin, **S. Wan**, Y. Kong, B. Mohapatra, H. Band, V. Band and B. Duan, "Embedded Bioprinting of Breast Tumor Cells and Organoids Using Low Concentration Collagen based Bioinks", *Advanced Healthcare Materials*, 2023, 2300905. (*JCR Q1; Rank: 8/96; IF: 10.0*)
20. J. Wang and **S. Wan***, "Editorial: Single Cell Meets Metabolism and Cancer Biology", *Frontiers in Oncology*, 2023, vol. 13, article 1125186. (*JCR Q2; Rank: 78/245; IF: 5.738*)
21. Y. Liu, J. Klein, R. Bajpai, Q. Tran, P. Kolekar, J. L. Smith, R. E. Ries, L. Dong, B. J. Huang, J. Wang, T. Alonzo, L. Tian, H. L. Mulder, K. Szlachta, T. I. Shaw, J. Ma, M. Walsh, G. Song, T. Westover, R. Autry, A. Gout, D. Wheeler, **S. Wan**, G. Wu, J. J. Yang, W. Evans, M. Loh, J. Easton, J. M. Klco, S. Meshinchi, P. A. Brown, S. M. Pruett-Miller and X. Ma, "Etiology of Oncogenic Fusions in 5,190 Childhood Cancers and Its Clinical and Therapeutic Implication", *Nature Communications*, 2023, vol. 14, no. 1739, pp. 1-18. (*JCR Q1; Rank: 4/72; IF: 14.919*)
22. **S. Wan***, C. Jiang, S. Li and Y. Fan, "Special Issue on Bioinformatics and Machine Learning for Cancer Biology", *Biology*, 2022, vol. 11, no. 3, 361. (*JCR Q1; Rank: 16/93; IF: 5.079*)
23. T. Sakamoto, K. Batmanov, **S. Wan**, Y. Guo, L. Lai, R. B. Vega and D. P. Kelly, "The Nuclear Receptor ERR Cooperates with the Cardiogenic Factor GATA4 to Orchestrate Transcriptional Control of Cardiomyocyte Differentiation", *Nature Communications*, 2022, vol. 13, no. 1991, pp. 1-20. (*JCR Q1; Rank: 4/72; IF: 14.919*)
24. **S. Wan*** and J. Wang*, "A Sequence Obfuscation Method for Protecting Personal Genomic Privacy", *Frontiers in Genetics*, 2022, vol. 13, article 876686. (*JCR Q2; Rank: 48/176; IF: 4.599*)

Curriculum Vitae

25. R. Wang, X. Zheng, J. Wang, **S. Wan**, M. H. Wong, K. S. Leung and L. Cheng, “Improving Bulk RNA-seq Classification by Transferring Gene Signature from Single Cells in Acute Myeloid Leukemia”, *Briefings in Bioinformatics*, 2022, vol. 23, no. 2, bbac002. (**JCR Q1; Rank: 3/78; IF: 11.622**)
26. V. Honnell, J. Norrie, A. Patel, C. Ramirez, J. Zhang, K. Lai, **S. Wan** and M. A. Dyer, "Identification of a Modular Super-Enhancer in Murine Retinal Development", *Nature Communications*, 2022, vol. 13, no. 253, pp. 1-13. (**JCR Q1; Rank: 4/72; IF: 14.919**)
27. W. Qi, W. Rosikiewicz, Z. Yin, B. Xu, H. Jiang, **S. Wan**, Y. Fan, G. Wu and L. Wang, “Genomic Profiling Identifies Genes and Pathways Dysregulated by *HEY1-NCOA2* Fusion and Shed a Light on Mesenchymal Chondrosarcoma Tumorigenesis”, *Journal of Pathology*, 2022, vol. 257, no. 5, pp. 579-592. (**JCR Q1; Rank: 5/77; IF: 7.996**)
28. P. C. Chen, X. Han, T. Shaw, H. Sun, M. Niu, Z. Wang, Y. Jiao, B. Teubner, D. Eddins, L. Beloate, B. Bai, J. Mertz, Y. Li , Y. Fu , J. H. Cho , X. Wang , Z. Wu , S. Poudel , Z. F. Yuan, A. Mancieri, J. Low, H. M. Lee, M. Patton, L. Earls, E. Stewart, P. Vogel, **S. Wan**, G. Serrano, T. Beach, M. Dyer, R. Smeyne, T. Moldoveanu, T. Chen, G. Wu, S. Zakharenko, G. Yu and J. Peng, “Alzheimer’s Disease-Associated U1 snRNP Splicing Dysfunction Causes Neuronal Hyperexcitability and Cognitive Impairment”, *Nature Aging*, 2022, vol. 2, pp. 923-940. (**JCR Q1; Rank: 1/69; IF: 16.6**)
29. C. Jiang, **S. Wan**, P. Hu, Y. Li and S. Li, “Editorial: Transcriptional Regulation in Metabolism and Immunology”, *Frontiers in Genetics*, 2022, vol. 13, article 845697. (**JCR Q2; Rank: 48/176; IF: 4.599**)
30. **S. Wan***, J. Wang, G. Wu and Y. Fan*, “Robust and Ultrafast Clustering of Large-Scale Single Cell RNA-seq Data”, *Briefings in Bioinformatics*, 2022, submitted. (**JCR Q1; Rank: 3/78; IF: 11.622**)
31. S. Singh, W. Quarni, M. Goralski, **S. Wan**, H. Jin, L. A. Van de Velde, J. Fang, R. Sing, Y. Fan, M. Johnson, W. Akers, P. Murray, P. G. Thomas, D. Nijhawan, A. M. Davidoff and J. Yang, “Targeting the Spliceosome through RBM39 Degradation Results in Exceptional Responses in High-Risk Neuroblastoma Models”, *Science Advances*, 2021, vol. 7, no. 47, eabj5405. (**JCR Q1; Rank: 5/73; IF: 14.136**)
32. A. Lavado, R. Gangwar, J. Pare, **S. Wan**, Y. Fan and X. Cao, “YAP/TAZ Maintain the Proliferative Capacity and Structural Organization of Radial Glial Cells during Brain Development”, *Developmental Biology*, 2021, vol. 480, pp. 39-49. (**JCR Q2; Rank: 15/41; IF: 3.582**)
33. **S. Wan*** and J. Q. Wang, “KLNP-Chlo: Multi-Label Ensemble Kernelized Neighbourhood Propagation for Predicting Localization of Chloroplast Proteins”, *Bioinformatics*, 2021, submitted. (**JCR Q1; Rank: 3/58; IF: 6.937**)
34. **S. Wan***, J. Z. Hu and J. Q. Wang, “Improving Prediction of Multi-Location Protein Subcellular Localization with Confidence-Weighted Gene Ontology Features”, *Bioinformatics*, 2021, submitted. (**JCR Q1; Rank: 3/58; IF: 6.937**)
35. M. Zanin, S. S. Wong, P. Schreiner, **S. Wan**, P. Vogel, Z. Cheng, R. El-Shesheny, D. F. Boyd, J. C. Jones, J. DeBeauchamp, Z. A. Kocer, P. Thomas, S. Barman, T. C. Chang, Y. Fan, R. Webster, R. Webby, “Airborne Transmission of Influenza Viruses is Associated with a Distinct Host Transcriptomic Response in the Nasal Epithelium”, *Proceedings of the National Academy of Sciences of the United States of America (PNAS)*, 2021, submitted. (**JCR Q1; Rank: 8/73; IF: 11.205**)

36. C. Jablonowski, G. Wu, W. Quarni, H. Hu, S. Singh, D. Bostanthirige, H. Jin, T. C. Chang, **S. Wan**, D. Finkelstein, J. H. Cho, D. Hu, V. Pagala, S. Miller, R. Wang, A. Murphy, K. Freeman, A. B. Shyu, A. M. Davidoff, J. Peng, J. Yang, “Metabolic Reprogramming of Cancer Cells by JMJD6-Mediated Alternative Splicing”, *Science Advances*, 2021, submitted. (*JCR Q1; Rank: 5/73; IF: 14.136*)
37. T. I. Shaw, Y. Li, H. Wang, A. Gout, X. Von Buttlar, **S. Wan**, X. Ma, A. C. Tan, “Analytical Challenges of Alternative Splicing in Cancer”, *Frontiers in Oncology*, 2021, submitted. (*JCR Q2; Rank: 62/242; IF: 6.244*)
38. **S. Wan**, J. Kim and K. J. Won, “SHARP: Hyper-Fast and Accurate Processing of Single-Cell RNA-seq Data via Ensemble Random Projection”, *Genome Research*, 2020, vol. 30, pp. 205-213. (*JCR Q1; Rank: 6/177; IF: 11.093*)
39. T. Sakamoto, T. Matsuura, **S. Wan**, D. Ryba, J. Kim, K. J. Won, L. Lai, C. Petucci, N. Petrenko, K. Musunuru, R. Vega, D. Kelly, “A Critical Role for Estrogen-Related Receptor Signaling in Cardiac Maturation”, *Circulation Research*, 2020, vol. 126, pp. 1685-1702. (*JCR Q1; Rank: 5/138; IF: 14.467*)
40. B. Ahn, **S. Wan**, N. Jaiswal, R. Vega, D. E. Ayer, P. M. Titchenell, X. Han, K. J. Won, and D. P. Kelly, “MondoA Coordinately Drives Muscle Lipid Accumulation and Insulin Resistance”, *JCI Insight*, 2019, 4(15): e129119. (*JCR Q1; Rank: 14/138; IF: 6.205*)
41. **S. Wan*** and M. W. Mak*, “Predicting Subcellular Localization of Multi-Location Proteins by Improving Support Vector Machines with Adaptive-Decision Schemes”, *International Journal of Machine Learning and Cybernetics*, 2018, vol. 9, pp. 399–411. (*JCR Q1; Rank: 31/134; IF: 3.844*)
42. **S. Wan***, M. W. Mak*, and S. Y. Kung, "FUEL-mLoc: Feature-Unified Prediction and Explanation of Multi-Localization of Cellular Proteins in Multiple Organisms", *Bioinformatics*, 2017, vol. 33, no. 5, pp. 749–750. (*JCR Q1; Rank: 3/58; IF: 6.937*)
43. **S. Wan***, M. W. Mak*, and S. Y. Kung, “Transductive Learning for Multi-Label Protein Subchloroplast Localization Prediction”, *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2017, vol. 14, pp. 212–224. (*JCR Q1; Rank: 15/125; IF: 3.710*)
44. **S. Wan***, M. W. Mak*, and S. Y. Kung, "Gram-LocEN: Interpretable Prediction of Multi-Location Gram-Positive and Gram-Negative Bacterial Protein Subcellular Localization ", *Chemometrics and Intelligent Laboratory Systems*, 2017, vol. 162, pp. 1–9. (*JCR Q1; Rank: 11/123; IF: 2.895*)
45. J. Q. Wang, C. C. Zhang, **S. Wan** and G. Peng, "Is Congenital Amusia a Connectome Disorder?: A Diffusion MRI Study Combining Tract- and Network-Based Analysis", *Frontiers in Human Neurosciences*, 2017, vol. 11, pp. 473. doi: 10.3389/fnhum.2017.00473. (*JCR Q2; Rank: 24/77; IF: 2.673*)
46. **S. Wan***, M. W. Mak*, and S. Y. Kung, "Ensemble Linear Neighbourhood Propagation for Predicting Subchloroplast Localization of Multi-Location Proteins", *Journal of Proteome Research*, 2016, vol. 15, pp. 4755–4762. (*JCR Q1; Rank: 17/78; IF: 4.466*)
47. **S. Wan***, M. W. Mak*, and S. Y. Kung, “Sparse Regressions for Predicting and Interpreting Subcellular Localization of Multi-Label Proteins”, *BMC Bioinformatics*, 2016, 17:97. (*JCR Q1; Rank: 9/59; IF: 3.242*)

Curriculum Vitae

48. **S. Wan***, M. W. Mak*, and S. Y. Kung, "Mem-mEN: Predicting Multi-Functional Types of Membrane Proteins by Interpretable Elastic Nets", *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2016, vol. 13, pp. 706–718. (**JCR Q1; Rank: 15/125; IF: 3.710**)
49. **S. Wan***, M. W. Mak*, and S. Y. Kung, "Benchmark Data for Identifying Multi-Functional Types of Membrane Proteins", *Data in Brief*, 2016, vol. 8, pp. 105–107. (**JCR Q2; Rank: 58/128; IF: 1.2**)
50. **S. Wan***, M. W. Mak*, and S. Y. Kung, "Mem-ADSVM: A Two-Layer Multi-Label Predictor for Identifying Multi-Functional Types of Membrane Proteins", *Journal of Theoretical Biology*, 2016, vol. 398, pp. 32–42. (**JCR Q1; Rank: 13/57; IF: 2.113**)
51. **S. Wan***, M. W. Mak, and S. Y. Kung, "mLASSO-Hum: A LASSO-Based Interpretable Human-Protein Subcellular Localization Predictor", *Journal of Theoretical Biology*, 2015, vol. 382, pp. 223–234. (**JCR Q1; Rank: 13/57; IF: 2.113**)
52. **S. Wan**, M. W. Mak, and S. Y. Kung, "mPLR-Loc: An Adaptive-Decision Multi-Label Classifier Based on Penalized Logistic Regression for Protein Subcellular Localization Prediction", *Analytical Biochemistry*, 2015, vol. 473, pp. 14–27. (**JCR Q2; Rank: 33/86; IF: 2.877**)
53. **S. Wan**, M. W. Mak, and S.Y. Kung, "HybridGO-Loc: Mining Hybrid Features on Gene Ontology for Predicting Subcellular Localization of Multi-Location Proteins", *PLoS ONE*, 2014, 9(3): e89545. (**JCR Q1; Rank: 15/64; IF: 2.806**)
54. **S. Wan**, M. W. Mak, and S. Y. Kung, "R3P-Loc: A Compact Multi-Label Predictor Using Ridge Regression and Random Projection for Protein Subcellular Localization", *Journal of Theoretical Biology*, 2014, vol. 360, pp. 34–45. (**JCR Q1; Rank: 13/57; IF: 2.113**)
55. **S. Wan**, M. W. Mak, and S. Y. Kung, "Semantic Similarity over Gene Ontology for Multi-Label Protein Subcellular Localization", *Engineering*, 2013, vol. 5, pp. 68–72.
56. **S. Wan**, M. W. Mak, and S. Y. Kung, "GOASVM: A Subcellular Location Predictor by Incorporating Term-Frequency Gene Ontology into the General Form of Chou's Pseudo-Amino Acid Composition", *Journal of Theoretical Biology*, 2013, vol. 323, pp. 40–48. (**JCR Q1; Rank: 13/57; IF: 2.113**)
57. **S. Wan**, M. W. Mak, and S. Y. Kung, "mGOASVM: Multi-Label Protein Subcellular Localization Based on Gene Ontology and Support Vector Machines", *BMC Bioinformatics*, 2012, 13:290. **Highly accessed** (**JCR Q1; Rank: 9/59; IF: 3.242**)

CONFERENCE PAPERS

58. **S. Wan***, J. Kim, Y. Fan and K. J. Won*, "Hyper-Fast and Accurate Clustering of Ultra-Large-Scale Single-Cell Data with Ensemble Random Projection", *The 2020 International Conference on Machine Learning (ICML) Workshop on Computational Biology*, virtual online, Jul. 2020. (**Highlights**)

Curriculum Vitae

59. **S. Wan***, J. Kim, Y. Fan and K. J. Won*, “Processing millions of single cells by SHARP”, *The 11th ACM Conference on Bioinformatics, Computational Biology and Health Informatics (ACM BCB 2020)*, virtual online, Sep. 2020. (Highlights)
60. **S. Wan**, M. W. Mak and S. Y. Kung, "Protecting Genomic Privacy by a Sequence-Based Obfuscation Method", 2017, arXiv preprint, arXiv: 1708.02629.
61. Mert Al, **S. Wan** and S. Y. Kung, "Ratio Utility and Cost Analysis for Privacy Preserving Subspace Projection", 2017, arXiv preprint, arXiv:1702.07976.
62. **S. Wan**, M. W. Mak, B. Zhang, Y. Wang and S. Y. Kung, "Ensemble Random Projection for Multi-Label Classification with Application to Protein Subcellular Localization", *2014 IEEE International Conference on Acoustic Speech and Signal Processing (ICASSP'14)*, Florence, Italy, May 2014, pp. 6040-6044.
63. **S. Wan**, M. W. Mak, B. Zhang, Y. Wang and S. Y. Kung, "An Ensemble Classifier with Random Projection for Predicting Multi-Label Protein Subcellular Localization", *The 2013 IEEE International Conference on Bioinformatics and Biomedicine (BIBM'2013)*, Shanghai, China, Dec. 2013, pp. 35-42.
64. **S. Wan**, M. W. Mak, and S. Y. Kung, "Adaptive Thresholding for Multi-Label SVM Classification with Application to Protein Subcellular Localization Prediction", *2013 IEEE International Conference on Acoustic Speech and Signal Processing (ICASSP'13)*, Vancouver, Canada, May 2013, pp. 3547-3551.
65. **S. Wan**, M. W. Mak, and S. Y. Kung, "GOASVM: Protein Subcellular Localization Prediction Based on Gene Ontology Annotation and SVM", *2012 IEEE International Conference on Acoustic Speech and Signal Processing (ICASSP'12)*, Kyoto, Japan, Mar. 2012, pp. 2229-2232.
66. **S. Wan**, M. W. Mak, and S. Y. Kung, "Protein Subcellular Localization Prediction Based on Profile Alignment and Gene Ontology", *2011 IEEE International Workshop on Machine Learning for Signal Processing (MLSP'11)*, Beijing, China, Sep. 2011, pp. 1-6.
67. **S. Wan**, C. Yao, Y. Hu, and G. Zhang, “A Method of Continuous Data Flow Embedded within Speech Signals”, *The 2-nd International Conference on Signal Acquisition and Processing (ICSAP'10)*, Bangalore, India, Feb. 2010, pp. 362-365.

CONFERENCE ABSTRACTS

68. M. Sun, J. Wang and **S. Wan***, “iS3RGs: Discriminating Diverse Medulloblastoma Subtypes by Leveraging Heterogenous Transcriptome Data with Batch Effects”, *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025.
69. L. Li, J. Khoury, J. Wang and **S. Wan***, “AttentionAML: An Attention-Based Deep Learning Model for Accurate Identification of Childhood Acute Myeloid Leukemia Subtypes”, *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025.
70. N. Peterson, M. Sun, J. Wang and **S. Wan***, “Identifying Pancreatic Cancer Subtypes by a Novel Meta-Learning Model”, *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025.
71. X. Wu, J. Wang and **S. Wan***, “Combining Random Projection and Stacking Learning for NSCLC Subtype Classification Based on Transcriptomic Data”, *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025.

Curriculum Vitae

72. M. J. Baek, L. Li, J. Wang, V. Band and S. Wan*, “A Weighted Multi-Modal Transfer Learning Model for Alleviating Racial Disparities in Breast Cancer”, *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025.
73. M. Sun, J. Wang and S. Wan*, “Identifying Antimicrobial Peptides Based on Proportionalized Split Amino Acid Compositions”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
74. M. Sun, J. Wang and S. Wan*, “Discriminating Diverse Medulloblastoma Subtypes by Leveraging Heterogenous Transcriptome Data with Batch Effects”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
75. L. Li, J. Khoury, J. Wang and S. Wan*, “Pediatric Acute Myeloid Leukemia Classification via Multi-Modal Ensemble Learning”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
76. H. Xiao, J. Wang and S. Wan*, “WIMOAD: Weighted Integration of Multi-Omics Data for Alzheimer’s Disease (AD) Diagnosis”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
77. M. Sun, J. Wang and S. Wan*, “iS3RGs: Discriminating Diverse Medulloblastoma Subtypes by Leveraging Heterogenous Transcriptome Data with Batch Effects”, *Pediatric Cancer Research Symposium*, Omaha, NE, Aug. 2024.
78. L. Li, J. Khoury, J. Wang and S. Wan*, “AttentionAML: An Attention-Based Deep Learning Model for Identifying Acute Myeloid Leukemia Subtypes”, *Pediatric Cancer Research Symposium*, Omaha, NE, Aug. 2024.
79. L. Li, H. Xiao, J. Khoury, J. Wang and S. Wan*, “RanBALL: Identifying B-Cell Acute Lymphoblastic Leukemia Subtypes Based on an Ensemble Random Projection Model”, *Cancer Research*, 2024, vol. 84 (6_Supplement), pp. 4907-4907. (presented in *AACR Annual Meeting 2024*, San Diego, CA, Apr. 2024)
80. L. Li, J. Wang and S. Wan*, “Reducing Health Disparities for Prostate Adenocarcinoma by Integrating Multi-Omics Data via a Multi-Modal Transfer Learning Approach”, *Cancer Research*, 2024, vol. 84 (6_Supplement), pp. 4800-4800. (presented in *AACR Annual Meeting 2024*, San Diego, CA, Apr. 2024)
81. L. Li, H. Xiao and S. Wan*, “B-Cell Acute Lymphoblastic Leukemia Subtype Identification with an Ensemble Random Projection-Based Machine Learning Model”, *CHRI Scientific Conference*, Omaha, NE, Nov. 2023.
82. L. Li and S. Wan*, “Integrating Multi-Omics Data by a Multi-Modal Transfer Learning Model to Reduce Healthcare Disparities for Kidney Renal Clear Cell Carcinoma”, *CHRI Scientific Conference*, Omaha, NE, Nov. 2023.
83. J. Feng, M. Sun, W. Zhang, G. Wang and S. Wan*, “SAMP: An Accurate Ensemble Model Based on Proportionalized Split Amino Acid Composition for Identifying Antimicrobial Peptides”, *Antimicrobial Peptides: Yesterday, Today and Tomorrow*, Omaha, NE, Oct. 2023.
84. L. Li, H. Xiao and S. Wan*, “RanBALL: an Ensemble Random Projection-Based Model for Identifying B-Cell Acute Lymphoblastic Leukemia Subtypes”, *PCRG Symposium 2023*, Omaha, NE, Aug. 2023.

Curriculum Vitae

85. W. Qi, W. Rosikiewicz, Z. Yin, B. Xu, **S. Wan**, Y. Fan, G. Wu and L. Wang, “RNA-seq and ChIP-seq profiling identifies genes and pathways dysregulated by HEY1-NCOA2 fusion and shed a light on mesenchymal chondrosarcoma tumorigenesis”, *Cancer Research*, 2021, vol. 81 (13_Supplement), pp. LB188-LB188. (presented in *AACR Annual Meeting 2021*, Philadelphia, PA, Apr. 2021)
86. T. Sakamoto, **S. Wan**, K. Batmanov, and D. P. Kelly, "The Estrogen-Related Receptor (ERR) Drives Cardiac Myocyte Maturation in Cooperation With GATA4", *Circulation Research*, 2020, vol. 127 (Suppl_1), pp. A222-A222.
87. **S. Wan**, J. Kim, K. J. Won, and Y. Fan, “Hyper-Fast and Accurate Clustering of Ultra-Large-Scale Single-Cell Data with Ensemble Random Projection”, *Cell Symposia: The Conceptual Power of Single-Cell Biology*, San Francisco, CA, USA, Apr. 2020.
88. T. Sakamoto, **S. Wan**, K. J. Won, and D. P. Kelly, “The Estrogen-Related Receptor Coordinates Transcription of Genes Involved in Mitochondrial and Contractile Maturation in Human Induced Pluripotent Stem Cell-Derived Cardiac Myocytes”, *Circulation*, 2019, vol. 140 (Suppl_1), pp. A11803-11803. (presented in *American Heart Association Scientific Sessions (AHA2019)*, Philadelphia, PA, USA, Nov. 2019)
89. T. R. Matsuura, T. Sakamoto, D. M. Ryba, **S. Wan**, and D. P. Kelly, “Estrogen-Related Receptor Signaling Is Critical for Postnatal Cardiac Maturation”, *Circulation*, 2019, vol. 140 (Suppl_1), A13898-A13898. (presented in *American Heart Association Scientific Sessions (AHA2019)*, Philadelphia, PA, USA, Nov. 2019)
90. B. Ahn, **S. Wan**, K. J. Won, N. Jaiswal, P. M. Titchenell, and D. P. Kelly, “MondoA Mediates Myocyte Lipid Accumulation and Insulin Resistance Driven by Chronic Nutrient Excess”, *American Diabetes Association's 79th Scientific Sessions*, San Francisco, CA, USA, Jun. 2019. (oral)
91. **S. Wan**, J. Kim and K. J. Won, "Hyper-Fast and Accurate Processing of Large-Scale Single-Cell Transcriptomics Data via Ensemble Random Projection", *RECOMB/ISCB Conference on Regulatory & Systems Genomics with DREAM Challenges (RSG DREAM 2018)*, New York City, NY, USA, Dec. 2018.

GRANTS

1. **Number:** 1R03OD038391-01 (Wan)

Role: PI

Sponsor: DHHS/NIH/Office of The Director

Title: Leveraging Heterogeneous Common Fund Data Sets and Beyond for Identifying Lung Cancer Subtypes

Dates: 09/05/2024 – 09/04/2025

Amount: \$307,000 (Dr. Wan's portion: \$307,000)

Effort: 3.6 calendar months

Status: active

2. **Number:** 3P30CA036727-37S5 (Sweasy)

Role: Project Leader/Point of Contact (POC)

Curriculum Vitae

Sponsor: DHHS/NIH/NCI (P30 CCSG Administrative Supplement)

Title: An Accurate Machine Learning Framework for Childhood Acute Myeloid Leukemia Subtype Identification by Integrating Bulk and Single-Cell Multi-Omics Data Within and Beyond the CCDI System

Dates: 09/26/2023 – 08/31/2025

Effort: 3.0 calendar month

Amount: \$500,000 (Dr. Wan's portion: \$500,000)

Status: active

3. Number: 3P20GM103427-23S2 (Sorgen)

Role: co-Project Leader

Sponsor: DHHS/NIH/NIGMS

Title: Nebraska Research Network in Functional Genomics

Dates: 06/20/2024 – 04/30/2025

Amount: \$958,437 (Dr. Wan's portion: \$216,668)

Effort: 3 calendar months

Status: active

4. Number: 31-3205-1243 (Wan)

Role: PI

Sponsor: Nebraska Research Initiative (NRI) Collaborative Grant

Title: An Artificial Intelligence Based Framework to Reduce Health Disparities for Breast Cancer

Dates: 07/01/2024 – 06/30/2025

Effort: 0.6 calendar month

Amount: \$100,000 (Dr. Wan's portion: \$100,000)

Status: active

5. Number: IRG-22-146-07-IRG (Wan)

Role: PI

Sponsor: American Cancer Society Institutional Research Grant (BCC-ACS-IRG) Seed Grants Programs

Title: Tackling Racial Disparities for Prostate Cancer by a Multi-Modal Transfer Learning Framework

Dates: 02/01/2024 – 01/31/2025

Effort: 0.6 calendar month

Amount: \$60,000 (Dr. Wan's portion: \$60,000)

Status: active

Curriculum Vitae

6. Number: 31-7210-6586 (Wan)

Role: PI

Sponsor: BCC-CHRI-PCRG Pilot Projects Program

Title: Advanced Machine Learning for Identifying Subtypes of B-Cell Acute Lymphoblastic Leukemia Based on Transcriptomic Profiling

Dates: 02/01/2023 – 01/31/2025

Effort: 1.0 calendar month

Amount: \$50,000 (Dr. Wan's portion: \$50,000)

Status: active

7. Number: OIA-2044049 (Wan)

Role: PI

Sponsor: Nebraska EPSCoR FIRST Award

Title: A Unified Machine Learning Framework to Integrate Large-Scale Single Cell and Bulk Multi-Omics Data for Cancer Subtyping

Dates: 12/01/2023 – 11/30/2024

Effort: 0.6 calendar month

Amount: \$25,000 (Dr. Wan's portion: \$25,000)

Status: active

8. Number: 31-7210-6605 (Wan)

Role: PI

Sponsor: UNMC Child Health Research Institute (CHRI)

Title: An Integrated Multi-Modal Artificial Intelligence Approach for Categorizing Pediatric Acute Myeloid Leukemia

Dates: 07/01/2024 – 06/30/2025

Effort: 0.6 calendar month

Amount: \$50,000 (Dr. Wan's portion: \$50,000)

Status: active

9. Number: 1P50AA030407-01 Pilot Project (Wan)

Role: PI

Sponsor: Alcohol Center of Research-Nebraska (ACORN) Pilot Projects Program

Title: Machine Learning for Identifying Biomarkers Within and Between Brain Networks for Alcohol Use Disorder Diagnosis

Dates: 07/01/2023 – 06/30/2024

Curriculum Vitae

Effort: 0.6 calendar month

Amount: \$25,000 (Dr. Wan's portion: \$25,000)

Status: active

10. Number: 3P20GM130447-04S1 (Dunaevsky)

Role: co-I

Sponsor: DHHS/NIH/NIGMS

Title: Cognitive Neuroscience of Development and Aging Center Team Science Administrative Supplement

Dates: 08/01/2023 – 01/31/2025

Amount: \$939,079 (Dr. Wan's portion: 10% effort)

Effort: 1.2 calendar month

Status: active

11. Number: NA (Rizzo)

Role: Project Leader/Point of Contact (POC)

Sponsor: DHHS/NIH/NIGMS (Women's Health Supplement)

Title: Great Plains IDeA-CTR (Reducing Health Disparities for Breast Cancer by an Integrated Multi-Modal Transfer Learning Framework)

Dates: TBA

Amount: \$307,000 (Dr. Wan's portion: \$307,000)

Effort: 3 calendar months

Status: pending (received award emails, but waiting for the official NOA)

BIOINFORMATICS SOFTWARE TOOLS

- 1. RPSLearner:** Combining Random Projection and Stacking Learning for Lung Cancer Subtyping
<https://github.com/wan-mlab/RPSLearner>
- 2. RankBat:** A Gene-Ranking Approach to Identify Medulloblastoma Subtypes Based on Heterogeneous Transcriptomics Data with Severe Batch Effects
<https://github.com/wan-mlab/RankBat>
- 3. RanBALL:** An Ensemble Random Projection Model to Identify Subtypes of B-cell Acute Lymphoblastic Leukemia
<https://github.com/wan-mlab/RanBALL>
- 4. WIMOAD:** Weighted Integration of Multi-Omics data for Alzheimer's Disease (AD) Diagnosis
<https://github.com/wan-mlab/WIMOAD>
- 5. SAMP:** A Proportionalized Split Amino Acid Composition Based Model for Identifying Antimicrobial Peptides
<https://github.com/wan-mlab/SAMP>

Curriculum Vitae

6. **AttentionAML**: An Attention-based MLP Model for Identifying Acute Myeloid Leukemia Subtypes
<https://github.com/wan-mlab/AttentionAML>
7. **SHARP**: Single-cell RNA-seq Hyper-fast and Accurate processing via ensemble Random Projection
<https://github.com/shibiaowan/SHARP>
8. **IterMegaBLAST**: A Sequence-Similarity Based Obfuscation Method for Genomic Privacy Protection
<https://github.com/shibiaowan/IterMegaBLAST>
9. **FUEL-mLoc**: Feature-Unified Prediction and Explanation of Protein Multi-Localization of Cellular Proteins (for Eukaryote, Human, Plant, Gram-positive, Gram-negative and Virus)
<http://bioinfo.eie.polyu.edu.hk/FUEL-mLoc/>
10. **Gram-LocEN**: Interpretable Prediction of Subcellular Multi-Localization of Gram-Positive and Gram-Negative Bacterial Proteins (for Gram-positive Bacteria, Gram-negative Bacteria)
<http://bioinfo.eie.polyu.edu.hk/Gram-LocEN/>
11. **PolyU-Loc**: A Package of Web-Servers for Protein Subcellular Localization Prediction
http://bioinfo.eie.polyu.edu.hk/Book_website/
12. **EnTrans-Chlo**: Transductive Learning for Protein Subchloroplast Localization Prediction
<http://bioinfo.eie.polyu.edu.hk/EnTransChloServer/>
13. **LNP-Chlo**: Linear Neighborhood Propagation for Protein Subchloroplast Localization Prediction
<http://bioinfo.eie.polyu.edu.hk/LNPChloServer/>
14. **Mem-ADSVM**: A Two-Layer Predictor for Multi-Label Membrane Protein Type Prediction
<http://bioinfo.eie.polyu.edu.hk/MemADSVMServer/>
15. **Mem-mEN**: A Predictor for Membrane Protein Multi-Functional Type Prediction
<http://bioinfo.eie.polyu.edu.hk/MemmENServer/>
16. **SpaPredictor**: An Interface of Two Predictors for Interpretable Protein Subcellular Localization
<http://bioinfo.eie.polyu.edu.hk/SpaPredictorServer/>
17. **HybridGO-Loc**: Mining Hybrid GO Features for Protein Subcellular Localization Prediction (for Virus and Plant)
<http://bioinfo.eie.polyu.edu.hk/HybridGoServer/>
18. **R3P-Loc**: Compact Predictor for Multi-Label Protein Subcellular Localization (for Eukaryote and Plant)
<http://bioinfo.eie.polyu.edu.hk/R3PLocServer/>
19. **mPLR-Loc**: Probabilistic Predictor for Multi-label Protein Subcellular Localization (for Virus and Plant)
<http://bioinfo.eie.polyu.edu.hk/mPLRLocServer/>
20. **mLASSO-Hum**: An Interpretable Predictor for Human Protein Subcellular Localization
<http://bioinfo.eie.polyu.edu.hk/mLASSOHumServer/>
21. **mGOASVM**: Multi-label Protein Subcellular Localization Prediction (for Virus and Plant)
<http://bioinfo.eie.polyu.edu.hk/mGoaSvmServer/mGOASVM.html>

22. GOASVM: Single-label Protein Subcellular Localization Prediction (for Eukaryote and Human)

<http://bioinfo.eie.polyu.edu.hk/mGoaSvmServer/GOASVM.html>

INVITED TALKS

1. TBA, in *Wyoming Sensory Biology Center of Biomedical Research Excellence (WSBC)* and *Department of Zoology and Physiology of University of Wyoming*, Laramie, WY, USA, April 2025.
2. “Advancing Biomedical Research by Artificial Intelligence and Bioinformatics”, invited as a **Featured Speaker** for the **1st Annual Basic Biomedical Research Retreat** of **University of Nebraska Medical Center**, Omaha, NE, USA, March 2025.
3. “Machine Learning for Large-Scale Single Cell Analysis”, in *Immunology Interest Group* of **University of Nebraska Medical Center**, Omaha, NE, USA, February 2025.
4. “Artificial Intelligence for Omics-Based Biomedical Research”, invited as a **Featured Speaker** for **The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)**, Houston, TX, USA, October 2024.
5. “Analyzing Millions of Single Cells via SHARP”, invited as a **Featured Speaker** for **Bioinformatics Research Symposium and Workshop on Integrative Omics and ML**, Lincoln, NE, USA, August 2024.
6. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Center for Aquacultural Breeding Research* of **Shanghai Ocean University**, Shanghai, China, July 2024.
7. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Department of Electrical and Electronic Engineering* of **The Hong Kong Polytechnic University**, Hong Kong SAR, China, July 2024.
8. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Health Data Science Center* of **Shenzhen People’s Hospital**, Shenzhen, China, July 2024.
9. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Institute of Artificial Intelligence* of **Wuhan University**, Wuhan, China, June 2024.
10. “Artificial Intelligence for Omics-Based Biomedical Research”, in *School of Artificial Intelligence and Automation* of **Huazhong University of Science and Technology**, Wuhan, China, June 2024.
11. “RanBALL: A Machine Learning Model Based on Ensemble Random Projection for Accurate Identification of B-Cell Acute Lymphoblastic Leukemia Subtypes”, in *Hematological Malignancies Group* of **University of Nebraska Medical Center**, Omaha, NE, USA, June 2024.
12. “Artificial Intelligence for Omics-Based Biomedical Research”, invited as a **Featured Speaker** for the Symposium on **ML/AI Applications in Biology and Medicine**, Omaha, NE, USA, May 2024.
13. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Department of Animal Sciences* of **University of Nebraska-Lincoln**, Lincoln, NE, USA, March 2024.
14. “Machine Learning for Omics-Based Biomedical Research”, in *VA Research Seminar* of **VA Nebraska-Western Iowa Health Care System**, Omaha, NE, USA, January 2024.

Curriculum Vitae

15. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Surgery Research Forum of Department of Regenerative Medicine of **University of Nebraska Medical Center***, Omaha, NE, USA, January 2024.
16. “Machine Learning for Processing Large-Scale Biological Data”, invited as a **Keynote Speaker** for *The 2023 Midlands Society of Physiological Sciences (MSPS) Annual Meeting*, Omaha, NE, USA, October 2023.
17. “Machine Learning for Large-Scale Bioinformatics Analysis”, in *Department of Biochemistry and Molecular Biology of **University of Nebraska Medical Center***, Omaha, NE, USA, June 2023.
18. “Machine Learning for Processing Large-Scale Biological Data”, in *Department of Biostatistics of **University of Nebraska Medical Center***, Omaha, NE, USA, January 2023.
19. “Machine Learning for Processing Large-Scale Biological Data”, in *Department of Genetics, Cell Biology and Anatomy of **University of Nebraska Medical Center***, Omaha, NE, USA, May 2022.
20. “Machine Learning for Processing Large-Scale Biological Data”, in *Department of Chemistry and Chemical Biology of **Stevens Institute of Technology***, Hoboken, NJ, USA, April 2022.
21. “Machine Learning and Data Science for Biomedicine and Cancer Research”, in ***BNU-HKBU United International College***, Zhuhai, China, April 2022.
22. “Machine Learning for Large-Scale Single-Cell Data Processing”, in *Department of Chemical Biology and Therapeutics of **St. Jude Children’s Research Hospital***, Memphis, TN, USA, February 2021.
23. “Supervised Learning”, in *Center for Applied Bioinformatics of **St. Jude Children’s Research Hospital***, Memphis, TN, USA, January 2021.
24. “Introduction to Machine Learning”, in *Center for Applied Bioinformatics of **St. Jude Children’s Research Hospital***, Memphis, TN, USA, November 2020.
25. “SHARP: Single-Cell RNA-seq Hyper-Fast and Accurate Clustering via Ensemble Random Projection”, in *School of Computer Science and Technology of **Xidian University***, Xi’an, China, August 2020.
26. “Assessing Gene Expression Dynamics with SLAM-Seq”, in *Center for Applied Bioinformatics of **St. Jude Children’s Research Hospital***, Memphis, TN, USA, May 2020.
27. “SHARP: Hyper-fast and Accurate Processing of Single-Cell RNA-Seq Data via Ensemble Random Projection”, in *Center for Applied Bioinformatics of **St. Jude Children’s Research Hospital***, Memphis, TN, USA, February 2020.
28. “SHARP: Single-Cell RNA-Seq Hyper-fast and Accurate Processing via Ensemble Random Projection”, in *Single-Cell Working Group at Smilow Center for Translational Research of **University of Pennsylvania***, Philadelphia, PA, USA, February 2019.
29. “Machine Learning for Bioinformatics Data Science”, in *Department of Computational Biology of **St. Jude Children’s Research Hospital***, Memphis, TN, USA, January 2019.
30. “Machine Learning for Bioinformatics Data Science: A Study on Protein Subcellular Localization”, in *Department of Electronic Engineering of **City University of Hong Kong***, Hong Kong SAR, China, June 2017.
31. “Machine Learning for Predicting Protein Subchloroplast Localization”, in *Department of Computer Science of **University of Bristol***, Bristol, UK, June 2017.

Curriculum Vitae

32. “Data Science in Bioinformatics: A Study on Protein Subcellular Localization”, in *Center for Big Data Research of Tsinghua-Berkeley Shenzhen Institute*, Shenzhen, China, May 2017.
33. “Predicting Protein Subcellular Localization by Bioinformatics Approaches”, in *Department of Pediatrics of The Children's Hospital of Philadelphia*, Philadelphia, PA, USA, April 2017.
34. “Combining Feature-Selection and Subspace Approaches for Utility-Privacy Tradeoff”, in *Department of Electrical Engineering of Princeton University*, Princeton, NJ, USA, March 2017.
35. “Bioinformatics Approaches for Protein Subcellular Localization Prediction”, in *Institute for Diabetes, Obesity and Metabolism of Perelman School of Medicine at University of Pennsylvania*, Philadelphia, PA, USA, March 2017.
36. “Linear and Kernelized Neighborhood Propagation: A Semi-Supervised Approach”, in *Department of Electrical Engineering of Princeton University*, Princeton, NJ, USA, December 2016.
37. “Semi-Supervised Multi-Kernel Learning in Image Classification”, in *Department of Electrical Engineering of Princeton University*, Princeton, NJ, USA, October 2016.
38. “Large-Scale in-Silico Prediction of Protein Subcellular Localization”, in *the 3-rd Wuhan University International Forum for Interdisciplinary Sciences and Engineering*, in *School of Electronic Information of Wuhan University*, Wuhan, China, April 2016.
39. “Machine Learning for Multi-Label Protein Subcellular Localization”, in *Department of Computer Science of Hong Kong Baptist University*, Hong Kong SAR, China, May 2015.
40. “Machine Learning Approaches for Protein Subcellular Localization”, in *Department of Statistics of The Chinese University of Hong Kong*, Hong Kong SAR, China, Apr. 2015.
41. “CPTAC Sub-Project: Customized Sample-Specific Protein Sequence Database”, in *Computational Bioinformatics and Bioimaging Laboratory (CBIL) of Virginia Tech*, VA, USA, May 2013.
42. “mGOASVM: A Multi-Label Protein Subcellular Localization Predictor”, in *Computational Bioinformatics and Bioimaging Laboratory (CBIL) of Virginia Tech*, VA, USA, May 2013.
43. “Semantic Similarity over Gene Ontology for Multi-label Protein Subcellular Localization”, in *2013 International Conference on Bioinformatics and Biomedical Engineering (iCBBE2013)*, Beijing, China, Sep. 2013.
44. “Protein Subcellular Localization Prediction Based on Profile Alignment and Gene Ontology”, in *the 6-th Beijing-Hong Kong International Doctoral Forum 2011*, Hong Kong SAR, China, Aug. 2011.
45. “Support Vector Data Description (SVDD) vs Support Vector Machine (SVM)”, in *Center for Multimedia Signal Processing of The Hong Kong Polytechnic University*, Hong Kong SAR, China, Mar. 2011.

AWARDS & HONORS

- **New Investigator Award**, awarded by UNMC, Omaha, NE, USA, 2024;
- **FIRST Award of Nebraska EPSCoR Program**, Lincoln, NE, USA, 2023;
- **IEEE Senior Member**, Piscataway, NJ, USA, 2022;
- **EIE Outstanding Young Alumni Award**, Hong Kong SAR, China, 2022;

Curriculum Vitae

- **Top reviewers in Cross-Field (top 1%)**, awarded by Clarivate, USA, 2019;
- **Top reviewers in Biology and Biochemistry (top 1%)**, awarded by Clarivate, USA, 2019;
- **Brandeis Postdoctoral Fellowship**, Princeton, NJ, USA, 2016-2018;
- **“Best Poster Award”** in the 6-th Beijing-Hong Kong International Doctoral Forum 2011, Hong Kong SAR, August, 2011;
- **Postgraduate Scholarship**, The Hong Kong Polytechnic University, Hong Kong SAR, 2010-2014;
- **National Scholarship**, Wuhan, China, 2008-2009;
- **National Inspirational Scholarship**, Wuhan, China, 2007-2008;
- **Excellent Scholarship** (the 3-rd prize), Wuhan, China, 2007-2008;
- **The Freshmen Scholarship**, Wuhan, China, 2006-2007.

TEACHING COURSES

- **2024**
 - *Tools and Algorithms in Bioinformatics* (BISB815), University of Nebraska Medical Center
 - *Modern Approaches in Cell Biology and Molecular Genetics* (MGCB912), University of Nebraska Medical Center
 - *Principles and Methodologies of Cancer Research* (CRGP880), University of Nebraska Medical Center
- **2023**
 - *Tools and Algorithms in Bioinformatics* (BISB815), University of Nebraska Medical Center
 - *Bioinformatics in Scientific Publications* (BISB903), University of Nebraska Medical Center
 - *Modern Approaches in Cell Biology and Molecular Genetics* (MGCB912), University of Nebraska Medical Center
 - *MGCB & BISB Fellowship Writing*, University of Nebraska Medical Center
- **2019-2022**
 - *Machine Learning for Applied Bioinformatics*, St Jude Children’s Research Hospital
 - *Journal Club for Machine Learning for Applied Bioinformatics*, St Jude Children’s Research Hospital
- **2017**
 - *Deep Neural Computing* (ELE571), Princeton University
- **2016**
 - *Kernel-Based Machine Learning* (ELE477), Princeton University
- **2015**
 - *Object-Oriented Design and Programming* (EIE320/EIE3375), The Hong Kong Polytechnic University
- **2014**
 - *Information Technology* (ENG2003), The Hong Kong Polytechnic University

Curriculum Vitae

- *Database Systems* (EIE3114), The Hong Kong Polytechnic University
- **2013**
 - *Information Technology* (ENG2003), The Hong Kong Polytechnic University
 - *Object-Oriented Design and Programming* (EIE320/EIE3375), The Hong Kong Polytechnic University
 - *Database Systems* (EIE3114), The Hong Kong Polytechnic University
 - *Distributed Systems and Network Programming* (EIE424/EIE4108), The Hong Kong Polytechnic University
- **2012**
 - *Object-Oriented Design and Programming* (EIE320/EIE3375), The Hong Kong Polytechnic University
 - *Distributed Systems and Network Programming* (EIE424/EIE4108), The Hong Kong Polytechnic University
 - *Communication Fundamentals* (EIE331), The Hong Kong Polytechnic University

MENTORED TRAINEES/POSTDOCS/STUDENTS

- University of Nebraska Medical Center
 - Graduate Students:
 - Mengtao Sun, PhD student
 - Lusheng Li, PhD student
 - Xinchao Wu, PhD student
 - Min-jeong Baek, PhD student
 - Kah Meng Soh, PhD student (co-supervised with Dr. Jieqiong Wang)
 - Hanyu Xiao, master student
 - Nicholas Peterson, master student
 - Yanan Lou, master student
 - Research Interns:
 - Junxi Feng, master student in Harvard University, USA
 - Andy Cong Liu, sophomore in University of Waterloo, Canada
 - Cece Zhang, freshman in University of Toronto, Canada
 - Yijin Zou, senior in China Agricultural University, China
 - Wenhao Zhang, master student in Xidian University, China
 - Xuehuan Zhu, master graduate in UCLA, USA
 - Simarpreet Kaur, BS-MS student in Indian Institute of Science Education and Research, India
 - Eswar Byrav Ramathurthy, freshman in University of Nebraska-Lincoln, USA
 - Grace (Jiaqi) Ge, freshman in Guangdong Technion-Israel Institute of Technology, China
 - Weiqi Ye, junior in University of California at Davis, USA

Curriculum Vitae

- Saiyi Cao, freshman in Guangdong Technion-Israel Institute of Technology, China
- Neil Linehan, master student in Western Governors University, USA
- Ouyang Han, master student in Northwestern University, USA
- Anulika Pearl Nwashili, master student in University of Arkansas at Little Rock, USA
- Sili Li, junior in University of Wisconsin – Madison, USA
- Navya Pavani Pachava, master student in University of Nebraska at Omaha, USA
- Princeton University
 - Graduate Advisee:
 - Mert Al (Supervisor: Prof. S. Y. Kung)
 - Undergraduate Advisee:
 - Artur Filipowicz (Supervisor: Prof. S. Y. Kung)

SERVICE WITHIN UNMC

- **Administration Committee**
 - Data Science Faculty Search Committee, College of Allied Health Profession of UNMC, 2025
 - AI Task Force, UNMC/Nebraska Medicine, 2023 - present
 - BISB Graduate Committee, UNMC, 2022 - present
- **Graduate Student Supervisory Committee**
 - Esther Ugwueke, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
 - Michelle Aguilar-Gaspar, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
 - Ruobin Liu, Biomedical Informatics (BMI) PhD Program, UNMC
 - Saadyeh Rashidi, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
 - Eric Zadok Mpingirika, Cancer Research (CR) PhD Program, UNMC
 - Ritesh Dey, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
 - Farshid Oruji, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
 - Ben Nolan, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
 - Ashish Das, Immunology, Pathology and Infectious Diseases (IPID) PhD Program, UNMC

PROFESSIONAL ACTIVITIES

- **Grant Reviewer** for
 - NIH Special Emphasis Panel (SEP) in Health Services and Systems Review Branch (HSS), 2025
 - NSF Graduate Research Fellowship Program (GRFP), 2024
 - Quebec Consortium for Drug Discovery (CQDM), Canada, 2024
 - UNMC College of Public Health Innovation Fund, 2024

Curriculum Vitae

- Poland National Science Center, Poland, 2024
- Medical Research Council (MRC) of UK Research and Innovation (UKRI), UK, 2024
- NIH RECOVER COVID Initiative Systems Biology Multiomics Study Section, 2024
- UNMC Graduate Studies Fellowship, 2024
- NIH Biodata Management and Analysis (BDMA) Study Section, 2023
- NSF Graduate Research Fellowship Program (GRFP), 2023
- NIH Early Career Reviewer (ECR) Program, 2023
- UNMC Buffet Cancer Center Pilot Grant Review Panel, 2023
- UNMC Graduate Studies Fellowship Study Section on Neuroscience, 2023
- UNMC Graduate Studies Fellowship Study Section on Community & Population Health/Nursing/Clinical & Translational Research, 2023
- UNMC Buffet Cancer Center Pilot Grant Review Panel, 2022
- **Editor-in-Chief** for *Current Proteomics* (SCI/JCR indexed)
- **Associate Editor** for *Heliyon* (**Impact Factor: 4; Cell Press/Elsevier**)
- **Editorial Board Member** for
 - *Briefings in Functional Genomics* (**Impact Factor: 4.840; Oxford University Press**)
 - *BMC Bioinformatics* (**Impact Factor: 3.328; Springer Nature**)
 - *Discover Oncology* (**Impact Factor: 2.8; Springer Nature**)
- **Academic Editor** for
 - *PeerJ Computer Science* (**Impact Factor: 3.8; PeerJ**)
 - *International Journal of Microbiology* (**Impact Factor: 3.4; Wiley**)
 - *BioMed Research International* (**Impact Factor: 3.411; Wiley**)
 - *Computational and Mathematical Methods in Medicine* (**Impact Factor: 2.809; Wiley**)
- **Guest Associate Editor** for
 - *International Journal of Molecular Sciences* (**Impact Factor: 5.6**) Research Topic on Ferroptosis: Emerging Mechanisms, Physiological Functions, and Therapeutic Applications
 - *Discover Life* (**Springer New Journal**)
 - *Frontiers in Psychology* (**Impact Factor: 3.8**) Research Topic on Subjective Well-Being and Human Decision Behaviors
 - *Frontiers in Immunology* (**Impact Factor: 8.786**) Research Topic on New Insight of Immunosenescence and Inflammaging in Geriatric Medicine and Multimorbidity
 - *Briefings in Functional Genomics* (**Impact Factor: 4.0**) Research Topic on Artificial Intelligence and Multi-Omics Research for Human Diseases
 - *BMC Methods* (**Springer New Journal**) Research Topic on Artificial Intelligence for Omics Data Analysis
 - *BMC Bioinformatics* (**Impact Factor: 3.328**) Research Topic on Evolutionary Studies

Curriculum Vitae

- *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Identification of Susceptibility Genes to Long COVID
- *Frontiers in Pharmacology* (**Impact Factor: 5.988**) Research Topic on Ferroptosis as a Novel Therapeutic Target for Inflammation-Related Diseases
- *Frontiers in Oncology* (**Impact Factor: 5.738**)
- *Biology* (**Impact Factor: 5.079**) Research Topic on Bioinformatics and Machine Learning for Cancer Biology (Volume II)
- *Genes* (**Impact Factor: 4.141**) Research Topic on Machine Learning Applications in Genetics
- *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Bioinformatics Analysis of Omics Data for Biomarker Identification in Clinical Research–Volume II
- *Frontiers in Cell and Developmental Biology* (**Impact Factor: 6.684**) Research Topic on Single Cell Meets Metabolism and Cancer Biology
- *Biology* (**Impact Factor: 5.079**) Research Topic on Bioinformatics and Machine Learning for Cancer Biology
- *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Transcriptional Regulation in Metabolism and Immunology
- *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Bioinformatics Analysis of Omics Data for Biomarker Identification in Clinical Research
- Editorial Board Member for *The Open Bioinformatics Journal*
- Editorial Board Member for *Gene & Translational Bioinformatics* (GTB)
- Topical Advisory Panel Member for *International Journal of Molecular Sciences* (**Impact Factor: 5.6**)
- Review Editor for *Frontiers in Immunology* (**Impact Factor: 8.786**)
- Review Editor for *Frontiers in Bioinformatics*
- **Co-Organizer** for the Symposium on **2024 ML/AI Applications in Biology and Medicine** in Omaha, NE.
- **TPC Member for International Conferences**
 - *The 2025 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology* (IAICT 2025);
 - *The 2025 IEEE International Conference on Internet of Things and Intelligence Systems* (IoTaIS 2025);
 - *The 36th IEEE International Conference on Tools with Artificial Intelligence* (IEEE ICTAI 2024);
 - *The 2024 IEEE International Conference on Internet of Things and Intelligence Systems* (IoTaIS 2024);
 - *The 5th International Conference on Smart Sensors and Application* (ICSSA 2024);
 - *The 10th International Conference on Mobile Applications* (MobiApps 2024);
 - *The 2024 IEEE Asia Pacific Conference on Wireless and Mobile* (APWiMob 2024);
 - *The 29th Asia-Pacific Conference on Communications* (APCC '2024);
 - *The 2024 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology* (IAICT 2024);

Curriculum Vitae

- *The 2024 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2024);*
- *The 35th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2023);*
- *The 9th International Workshop on Mobile Applications (MobiApps 2023);*
- *The 2023 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2023);*
- *The 2023 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2023);*
- *The 34th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2022);*
- *The 2022 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2022);*
- *The 2022 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2022);*
- *The 2022 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS 2022);*
- *The 2022 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT'2022);*
- *The 2022 International Conference on Decision Aid Sciences and Applications (DASA'2022);*
- *The 33rd IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2021);*
- *The 2021 IEEE International Conference on Internet of Things and Intelligent Systems (IoTaIS 2021);*
- *The 2021 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2021);*
- *The 2021 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2021);*
- *The 2021 International Conference on Data Analytics for Business and Industry (DATA'21);*
- *2021 International Conference on Innovation and Intelligence for Informatics, Computing and Technologies (3ICT 2021);*
- *The 32nd IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2020);*
- *The Second International Conference on Big Data and Advanced Wireless Technologies (BDAW 2020)*
- *The 1st Conference on Internet of Things and Embedded Intelligence (CITEI 2020)*
- *International Journal of Computing and Digital Systems'20 (IJCDS 2020)*
- *Bulletin of Electrical Engineering and Informatics (BEEI 2020);*
- *The 7th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2020);*
- *The 2nd Ahmad Dahlan International Conference Series on Engineering, Science and Information Technology (ADICS-ESIT 2020);*
- *The International Conference on Smart Education and Applied Social Sciences (SEAS 2020);*
- *The 31st IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2019);*
- *The 6th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2019);*

Curriculum Vitae

- *International Conference on Industry 4.0 and Artificial Intelligence Technologies (INAIT) 2019;*
- *International Conference on Digital Image and Signal Processing (DISP'2019) 2019;*
- *The 1st International Conference on Computer Science, Information Technology and Electrical Engineering (ICOMITEE'2019) 2019;*
- *Society of Engineering Science (SES'19) 2019;*
- *The 30th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2018);*
- *The First IEEE International Conference on Artificial Intelligence for Industries (AI4I 2018);*
- *2018 International Conference on Computer Intelligent Systems & Networking (ICCISN 2018);*
- *The 5th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2018);*
- *21st Saudi Computer Society National Computer Conference (SCS-NCC 2018);*
- *The 1st International Conference and Workshop on Telecommunication, Computing, Electronics and Control 2018 (ICW-TELKOMNIKA 2018);*
- *The 29th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2017);*
- *The 28th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2016);*
- **Reviewer for Journals**
 - *Nature Methods*
 - *Nature Communications*
 - *Nature Computational Science*
 - *IEEE Transactions on Neural Networks and Learning Systems (IEEE TNNLS)*
 - *Genome Medicine*
 - *Cancer Research*
 - *Nucleic Acids Research*
 - *British Journal of Cancer*
 - *Clinical and Translational Medicine*
 - *Journal of Nanobiotechnology*
 - *Bioinformatics*
 - *Briefings in Bioinformatics*
 - *Briefings in Functional Genomics*
 - *Journal of Proteome Research*
 - *PLoS Computational Biology*
 - *PLoS Genetics*
 - *PLoS One*
 - *Cancer Medicine*
 - *Knowledge-Based Systems*

Curriculum Vitae

- *Expert Systems with Applications*
- *iMeta*
- *Seminars in Cancer Biology*
- *Frontiers in Cell and Developmental Biology*
- *Frontiers in Immunology*
- *Frontiers in Oncology*
- *Frontiers in Microbiology*
- *Frontiers in Public Health*
- *Frontiers in Genetics*
- *Frontiers in Human Neuroscience*
- *BMC Bioinformatics*
- *Computational and Structural Biotechnology Journal*
- *Journal of Chemical Information and Modeling*
- *Cells*
- *Genomics*
- *Genes*
- *Proteomes*
- *Biochimie*
- *Biosensors*
- *Biotechnology Journal*
- *IEEE/ACM Transactions on Computational Biology and Bioinformatics (IEEE/ACM TCBB)*
- *IEEE Transactions on Nanobioscience (IEEE T-NB)*
- *IEEE Access*
- *Future Generation Computer Systems*
- *Journal of Theoretical Biology (JTB)*
- *Journal of Integrative Bioinformatics (JIB)*
- *Analytical Biochemistry (AB)*
- *Biology*
- *Micromachines*
- *Microorganisms*
- *Molecules*
- *Biomolecules*
- *Diagnostics*
- *Sensors*
- *BioMed Research International*

Curriculum Vitae

- *BMC Medical Informatics and Decision Making*
- *International Journal of Molecular Sciences (IJMS)*
- *International Journal of Machine Learning and Cybernetics (JMLC)*
- *Journal of Ambient Intelligence and Humanized Computing (AIHC)*
- *Symmetry*
- *Computational Biology and Chemistry*
- *Applied Mathematics and Computation (AMC)*
- *Journal of Applied Mathematics (JAM)*
- *Medicines*
- *Advances in Artificial Neural Systems (AANS)*
- *International Journal of Biomedical Imaging (IJBI)*
- *Computer Methods and Programs in Biomedicine (CMPB)*
- *International Journal of Disaster Risk Reduction (IJDRR)*
- *International Journal of Environmental Research and Public Health (IJERPH)*
- *Applied Sciences*
- *Electronics*
- *Cancers*

- **Reviewer for International Conferences**

- *2022 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2022); (8 times)*
- *2021 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2021); (8 times)*
- *The 2021 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2021); (2 times)*
- *The 2021 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2021); (1 time)*
- *2020 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2020); (7 times)*
- *The 1st Conference on Internet of Things and Embedded Intelligence (CITEI 2020); (2 times)*
- *Bulletin of Electrical Engineering and Informatics (BEEI 2020); (2 times)*
- *2019 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2019); (8 times)*
- *The 1st International Conference on Computer Science, Information Technology and Electrical Engineering (ICOMITEE'2019); (1 time)*
- *The 6th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2019); (1 time)*
- *2018 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2018); (3 times)*
- *The First IEEE International Conference on Artificial Intelligence for Industries (AI4I 2018); (2 times)*
- *The 5th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2018); (2 times)*

Curriculum Vitae

- 2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP2017); (3 times)
- 2017 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2017); (1 time)
- 2016 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2016); (3 times)
- 2015 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2015)–Workshop on Semantic Data Analytics and Bioinformatics (SDAB); (2 times)
- Registration coordinator for *The 8-th International Symposium on Chinese Spoken Language Processing (ISCSLP) 2012*
- Registration coordinator for *The 19-th International Conference on Digital Signal Processing (DSP) 2014*
- **Member**, Fred and Pamela Buffet Cancer Center (FPBCC) at UNMC
- **Member**, Children Health Research Institute (CHRI) at UNMC
- **Member**, Alcohol Center of Research: Nebraska (ACORN) at UNMC
- **Member**, Nebraska Center for Women’s Health Research (NCWHR) at UNMC
- **Member**, American Association of Cancer Research (AACR)
- **Member**, International Society for Computational Biology (ISCB)
- **Member**, Association for Computing Machinery (ACM)
- **Member**, Organization for Human Brain Mapping (OHBM)
- **Senior Member**, Institute of Electrical and Electronics Engineers (IEEE)